INDICATION: A 67-year-old female presenting with acute choledocholithiasis, right upper quadrant pain, conjugated hyperbilirubinemia (total bilirubin 5.8 mg/dL, direct 4.2 mg/dL), elevated alkaline phosphatase (380 U/L), and magnetic resonance cholangiopancreatography (MRCP) demonstrating a 9 mm stone in the distal common bile duct with upstream biliary dilatation.

PROCEDURE: Therapeutic Endoscopic Retrograde Cholangiopancreatography (ERCP), biliary sphincterotomy, mechanical balloon stone extraction, and temporary biliary stent placement.

NARRATIVE: The patient was positioned in the prone/left lateral decubitus position under general endotracheal anesthesia. An Olympus TJF-Q190V therapeutic duodenoscope was advanced into the descending duodenum. The major duodenal papilla appeared prominent and edematous without tumor invasion. Selective biliary cannulation was achieved through the major papilla using an Olympus V-System sphincterotome preloaded with a 0.035-inch Hydra Jagwire under fluoroscopic guidance.

Contrast injection (Omnipaque 300 diluted 50% with normal saline) demonstrated a dilated common bile duct measuring 12 mm in maximum luminal caliber. A single, discrete, well-defined 9 mm round radiolucent filling defect was visualized in the distal choledochus, along with smooth tapering into the ampulla. The intrahepatic biliary radicles were mildly dilated. A complete biliary sphincterotomy was executed along the 11 to 12 o'clock orientation using pure cut electrocautery (EndoCut mode) up to the transverse fold of the duodenal wall.

An extraction balloon was advanced over the guidewire above the filling defect, inflated to 11.5 mm, and withdrawn through the bile duct. The 9 mm calculus was successfully cleared into the duodenal lumen with a moderate release of biliary sludge and dark bile. Repeat occlusion cholangiogram confirmed complete ductal clearance without residual filling defects or extravasation. To ensure uninterrupted postoperative drainage and prevent secondary cholangitis, a 10-French x 7 cm Cotton-Leung plastic biliary stent was placed across the papilla into the common hepatic duct. Excellent bilious drainage was verified. The duodenoscope was withdrawn without complication.